Supplementary Table S5

Graph Neural Networks for PM2.5 Forecasting: Review on Recent Advancements, Challenges, and Outlooks

Table S5: Complete graph-representation and graph-construction evidence extracted from the 69 included studies. The fields distinguish the definition of graph nodes and edges, relation weighting and direction, graph type, graph information source, and construction procedure. Graph structures may use multiple simultaneous relation types; NR and unverified directions are retained where the primary extraction did not establish them.

Study	Graph node	Graph edge	Weighting	Direction	Graph type	Graph source	Construction details
S01 Wang et al. (2023)	Air-quality monitoring station; meteorological monitoring station (separate graphs)	Meteorological stations are connected when relative distance is <50 km; the PM2.5 prediction graph is reconstructed at each time point after meteorological-feature fusion	Unclear from full text	Meteorological graph: Undirected; PM2.5 prediction graph: NR	Multi-graph; static meteorological graph + time-varying PM2.5 graph	Distance; Meteorology; Dynamic; Hybrid	A meteorological graph is constructed among 9 meteorological stations using a 50 km distance criterion. GCN-LSTM extracts meteorological features, attention fuses meteorology with PM2.5/PM10 information, and a time-varying PM2.5 prediction graph is then constructed for 17 air-quality stations.
S02 Zhou et al. (2024)	Air-quality monitoring station	An initial geographical graph is built from inter-station distance; a learned spatiotemporal attention mechanism generates a dynamic graph used jointly with the initial graph	Weighted	Initial geographical graph: Undirected; learned dynamic graph: NR	Hybrid static + dynamic/adaptive graph	Distance; Learned; Adaptive; Dynamic; Hybrid	Initial adjacency uses geographical distance, with Aij = 1/dij when i≠j and dij<R, otherwise 0. Distance thresholds of 20 km, 60 km, and fully connected settings are evaluated. A bilinear spatiotemporal attention mechanism learns a dynamic spatial-dependency graph that is combined with the initial adjacency.
S03 Huo et al. (2026)	Prefecture-level city	Distance-graph edges connect city pairs whose Euclidean distance is <250 km; hyperedges connect three or more cities grouped by k-means clustering of historical pollutant series	Mixed: binary distance graph; hypergraph incidence structure	NR	Static hybrid graph; hypergraph	Distance; Correlation/statistical clustering; Hypergraph; Hybrid	Latitude/longitude are used to construct a pairwise distance graph; an edge exists when inter-city Euclidean distance is <250 km. A hypergraph is generated from the training data by k-means clustering of historical pollutant series. Cluster number selection uses the authors' stopping criterion $\Delta f<\tau$ with $\tau=0.05$.
S04 Jana et al. (2024)	Air-quality monitoring station	Stations are connected according to geographical distance using a distance threshold; the graph begins from pairwise station distances	Weighted — Gaussian distance-decay weighting	Undirected	Static physical/geographical graph	Distance; Physical/geographical topology	A distance-based graph is constructed over stations; edge weights use a Gaussian function of inter-station distance and a distance threshold is applied to determine connectivity.
S05 Qi et al. (2019)	Air-quality monitoring station	Stations are connected according to geographical distance within a 200 km threshold	Weighted — distance-based	Undirected	Static geographical graph	Distance; Physical/geographical topology	A geographical station graph is constructed using inter-station distance with threshold R = 200 km; spatial features are extracted by graph convolution before temporal modelling.
S06 Xiao et al. (2022)	City-level air-quality observation station / city	A static distance graph is combined with a time-varying directed wind graph; incoming and outgoing directed relationships are modelled separately	Weighted	Directed	Dynamic directed hybrid graph	Distance; Meteorology; Wind/transport; Dynamic; Hybrid	A static distance weight matrix is combined element-wise with a dynamic wind-based directed weight matrix. The model forms time-varying incoming and outgoing graphs and uses random-walk graph convolution along both directions.
S07 Sui and Han (2023)	Air-quality monitoring station	Two spatial views are constructed: a geographical-distance graph and a logical-similarity graph based on POI/road-network context	Weighted	Undirected	Static multi-view / multi-graph	Distance; Semantic/POI similarity; Multi-graph; Hybrid	The geographical graph uses distance-based weights. The logical graph uses cosine similarity of station-level POI/road-network feature vectors. Both views are processed by graph convolution and fused by attention.
S08 Teng et al. (2023)	Ground air-quality monitoring site	Dynamic directed transport edges connect monitoring sites according to distance, topographic blockage, wind direction, and wind speed	Weighted	Directed	Dynamic directed graph	Distance; Wind/transport; Topography; Dynamic; Hybrid	Candidate links are limited to ≤400 km and rejected when terrain blocks the path (terrain-height-difference criterion reported as >3 km). Wind creates directional source-to-receptor links; normalized transport weights are based on wind speed, angle, and inter-site distance and are updated for each prediction hour.
S09 Zhou et al. (2021)	Air-quality monitoring station	Time-varying directed links are determined by wind-field diffusion relationships among stations	Weighted — inverse wind-field diffusion distance	Directed	Dynamic directed graph	Wind/transport; Distance; Dynamic; Physical	A wind-field-based diffusion-distance surface is calculated and shortest transport paths are used to construct a dynamic directed graph. The reported edge weight is based on inverse wind-field diffusion distance.
S10 Zhou et al. (2022)	Air-quality monitoring station	Stations within 70 km are candidate neighbours; wind-field information determines time-varying directional relationships	Unclear from full text	Directed	Dynamic directed physics-guided graph	Distance; Wind/transport; Meteorology; Dynamic; Physics-guided	A neighbourhood is defined using a 70 km distance criterion. Wind direction/field information is then used to determine directed station relationships for each time window within a theory-guided graph-network framework.
S11 Chen et al. (2025)	City	Dynamic directed transport links are formed using geographical distance, wind direction/speed, travel distance, and terrain-obstruction criteria	Weighted — $W_{i,j} = 1 - dmin/L$ with L = 400 km	Directed	Dynamic directed hybrid graph	Distance; Wind/transport; Meteorology; Topography; Dynamic; Hybrid	A directed edge is retained only when the reported terrain, wind-angle, and transport-distance conditions are satisfied. The paper reports a 400 km transport limit and a terrain/elevation criterion; the elevation-difference wording is preserved as reported in the paper. Edge weight is $W_{i,j} = 1 - dmin/L$, L = 400 km.
S12 Guyu et al. (2025)	Air-quality monitoring station	Stations are connected using a distance graph based on Haversine distance and an inter-station threshold	Weighted — Aij = 1/dij for dij≤α and i≠j; otherwise 0	Undirected	Static distance graph	Distance; Physical/geographical topology	Haversine distance is calculated for every station pair. The adjacency weight is reciprocal distance when dij≤α; α is selected to cover 70% of pairwise inter-station distances.
S13 Shi et al. (2026)	Monitoring station (Beijing); city/spatial site (India)	Dual-scale graph: local geographical edges from distance kernel and thresholding; global all-pair learned attention; dynamic wind bias modifies spatial attention	Weighted	Mixed: static distance graph is symmetric; wind component introduces directional bias	Hybrid static local + adaptive global + dynamic wind-aware graph	Distance; Learned; Adaptive; Wind/transport; Dynamic; Hybrid	Local adjacency uses a Gaussian distance kernel exp(-(d/σ)^2), an adaptive local radius to avoid isolated nodes, and threshold τ = 0.1. Global all-pair attention learns long-range relationships. Wind direction is incorporated through u/v components and relative station direction to bias downwind interactions.
S14 Zhao et al. (2026)	Air-quality monitoring station	A multilayer air-quality network combines a spatial relationship matrix derived from physical distance/dispersion weighting and a temporal relationship matrix derived from pollutant time-series relationships	Weighted	NR	Multi-layer hybrid graph; spatial + temporal relationship network	Distance; Meteorology/wind; Correlation; Multi-graph/multilayer; Hybrid	The spatial relationship matrix uses Haversine distance and a Gaussian-plume-inspired weighting scheme; a temporal relationship matrix captures time-series relationships. These are combined into a spatiotemporal relationship matrix and multilayer network. Fuzzy Tsallis entropy is then used to estimate station importance.
S15 Ni et al. (2023)	Spatial region / monitoring location	A globally optimal static prior graph is combined with a learned local dynamic graph; candidate static graphs include TIN, geographic-threshold, KNN, and DTW constructions	Weighted	Mixed: initial prior graph is undirected; adaptive GAGCN is described as a weighted directed graph	Adaptive hybrid static + dynamic graph	Distance; DTW/statistical similarity; Learned; Adaptive; Dynamic; Hybrid	Static prior structures are tested using TIN, geographic-threshold, KNN, or DTW. LDGAT constructs local dynamic relationships from dot-product similarity of first-order temporal differences. GAGCN learns adaptive weights over graph information and supports scalable propagation.
S16 Kim et al. (2023)	PM2.5 monitoring station	A multi-graph represents wind-diffusion relationships and PM2.5 similarity relationships among stations	Weighted	Mixed: wind-diffusion graph is directed; similarity graph is symmetric	Multi-graph / hybrid transport + similarity graph	Wind/transport; Correlation/similarity; Multi-graph; Hybrid	A diffusion edge weight uses wind direction, wind speed, and station distance and is normalized with SoftMax. A separate PM2.5 similarity graph captures statistical similarity between station histories; both graph views are used by the spatiotemporal model.

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Supplementary Table S5 (continued)

Study	Graph node	Graph edge	Weighting	Direction	Graph type	Graph source	Construction details
S17 Zeng et al. (2025)	Air-quality monitoring station	Adjacency is generated adaptively from data by a learnable graph-generation layer and optimized during training	Weighted	NR	Adaptive / learned graph; model described as dynamic adaptive graph-generating	Learned; Adaptive; Distance (ablation prior)	The Adaptive Graph Generation (AGG) layer learns hidden spatial associations and continuously optimizes the adjacency matrix during model training. A predefined inverse-Euclidean-distance graph is used only in an ablation variant.
S18 Liu et al. (2026)	For the graph-integrated S-DI module, graph nodes are latent feature dimensions/components; monitoring-station spatial context is separately pooled by inverse distance	Time-varying soft interactions between latent dimensions are learned from the outer product of the latent vector and then transformed, sparsified, symmetrized, and normalized	Weighted	Undirected after explicit symmetrization	Learned dynamic latent graph	Learned; Dynamic	The Sparse Dynamic Interaction (S-DI) module forms Gt = zt zt^T, applies a diagonal bilinear transformation and nonlinearity, then sparsifies, symmetrizes, and normalizes the interaction matrix. The paper explicitly distinguishes this from a heavy conventional graph-neural module.
S19 Xu et al. (2023)	City-level observation station / city	A topology mask based on distance and altitude is combined with two edge-attribute graphs: a physical wind-advection graph and a learned adaptive edge-attribute graph	Weighted	Directed	Dynamic adaptive multi-graph / hybrid graph	Distance; Topography; Wind/transport; Learned; Adaptive; Dynamic; Hybrid	Topology P filters potential connections using a 300 km distance threshold and 1.2 km altitude threshold. One edge attribute is a wind-advection coefficient based on wind speed, direction, and distance; another edge attribute is learned adaptively by a neural network. Incoming and outgoing messages are processed separately.
S20 Ma et al. (2026)	Air-quality monitoring station	Four graph views are used: distance graph; binary neighborhood graph; pollutant-distribution similarity graph; functional/POI similarity graph	Mixed — distance, distribution-similarity and functional-similarity graphs are weighted; neighborhood graph is binary	NR	Static multi-graph with adaptive/dynamic graph-level ensemble weighting	Distance; Correlation/statistical distribution similarity; POI/functional similarity; Multi-graph; Adaptive/Learned; Hybrid	Distance graph uses an exponential distance kernel with threshold ξD; neighborhood graph is binary within distance δ; distribution-similarity graph uses Jensen–Shannon-based pollutant-distribution similarity with an exponential kernel and threshold ξS; functional graph uses cosine similarity of POI feature vectors with threshold ξF. Four graph-specific predictors are combined by the reinforcement-learning multi-graph controller (RLMC).
S21 Su et al. (2023)	Air-quality monitoring station	Two graph views: geographical-distance graph and POI-similarity graph	Weighted	NR	Static multi-graph	Distance; POI/functional similarity; Multi-graph; Hybrid	Distance graph uses inverse distance for station pairs within RD=15 km. POI graph uses cosine similarity of POI feature vectors with reported threshold RP=0.4. The two graph views are processed jointly by the spatial component.
S22 Zhao et al. (2025)	Air-quality monitoring station	A simple geographical graph connects stations within a distance threshold; a geographical hypergraph groups nearby stations; a clustering hypergraph groups stations using spectral clustering of historical PM2.5	Mixed — binary simple graph plus hypergraph incidence/attention weights	Undirected hypergraph; simple graph direction NR	Static hybrid simple graph + dual hypergraph	Distance; Statistical/spectral clustering; Hypergraph; Hybrid	The simple graph uses threshold m=20 km. A geographical hypergraph uses threshold interval (0,γ)=(0,10). A clustering hypergraph is generated by spectral clustering of historical PM2.5, with the number of cluster heads set to n=6.
S23 Gao et al. (2026)	Air-quality monitoring station	Two learned spatial structures are used: adaptive hypergraph incidence relationships and learned pairwise graph relationships	Weighted	NR	Adaptive learned graph + adaptive hypergraph; hybrid	Learned; Adaptive; Hypergraph; Hybrid	Hypergraph incidence H is learned from trainable node and hyperedge embeddings, passed through nonlinear transformations and top-k sparsification; the number of hyperedges is I=ρN and ρ=0.8. A learned pairwise adjacency is generated from node embeddings and top-k sparsification; k=20. In comparative experiments, predefined Pearson-correlation graphs and feature-cluster hypergraphs are used only as baselines/variants.
S24 Zhang et al. (2024)	City	Directed source–sink city relationships are allowed when city distance is within 300 km and intervening mountains are below 1200 m; edge attributes include source wind speed, inter-city distance, sink wind direction and an advection coefficient	Weighted — edge-channel mechanism recalculates edge importance from transport-related edge attributes	Directed	Directed physical graph with dynamic/time-varying edge attributes and learned edge weights	Distance; Wind/transport; Meteorology; Topography; Hybrid	Candidate transport edges are restricted by a 300 km distance threshold and 1200 m mountain-height criterion. Edge attribute S=ReLU(|v|cos(α)/d) represents advection using source wind speed |v|, sink wind direction α and source–sink distance d. ECGAT applies an edge-channel mechanism and GATConv to learn spatial relations.
S25 Mandal and Thakur (2023)	Air-quality monitoring station	Directed source-to-sink pollutant-transport relation using wind speed, wind direction and distance; spatial clustering groups stations before graph modelling	Weighted — transport/advection coefficient and graph-attention weights	Directed	Static clustered graph topology with time-varying meteorological edge features	Wind/transport; Distance; Meteorology; Statistical clustering; Hybrid	The directed edge feature is $P^t j \to i = \mathrm{ReLU}(\|v\|\cos(\theta)/D)$, where |v| is source-node wind speed, θ is the wind direction relative to the sink, and D is inter-station distance. Stations are grouped by clustering and processed with edge/node MLPs and GAT.
S26 Hu et al. (2024)	Air-quality monitoring station	Inter-station dependencies are represented by an end-to-end learned adaptive adjacency matrix; each hierarchical graph-convolution layer can learn a distinct spatial relationship structure	Weighted — learned adaptive adjacency values	NR	Adaptive; dynamic/hierarchical learned graph	Learned; Adaptive; Dynamic	The model learns source and target node representations and generates an adaptive adjacency end-to-end. A hierarchical mapping function updates graph relationships across graph-convolution layers. A predefined distance graph is used only in a comparison/ablation variant.
S27 Tan et al. (2022)	Monitoring site	A predefined adjacency matrix defines graph-neighbour relationships; GAT learns attention weights over neighbouring nodes. The criterion used to construct the adjacency matrix is NR.	Weighted — learned graph-attention coefficients	NR	Predefined graph with learned attention weights	Learned attention; adjacency source otherwise NR	The paper inputs an adjacency matrix to the GAT predictor and learns attention coefficients for neighbouring nodes; the actual rule used to construct the adjacency matrix is not reported.
S28 Gao and Li (2021)	Air-quality monitoring station	All station pairs are treated as candidate relationships through a trainable parameterized adjacency matrix Wadj; edge importance is optimized end-to-end with the forecasting task	Weighted — trainable parameterized adjacency matrix	Directed — the learned adjacency is asymmetric	Learned/adaptive parameterized directed graph	Learned; Adaptive	Rather than fixing a distance-based graph, the main GLSTM makes Wadj a trainable parameter matrix and optimizes it jointly with the forecasting model. Unimportant station-to-station relationships can be driven toward zero. A distance-based D-GLSTM is evaluated only as an ablation/comparison variant.
S29 Liu et al. (2021)	Monitoring site/node	Pearson correlation is computed between PM2.5 series. If correlation <0.5, the edge weight is set to 0; if correlation >0.5, reciprocal spatial distance is used as the connection value; self-connection value is 1.	Weighted — correlation threshold plus reciprocal-distance weights	NR	Static hybrid correlation–distance graph	Correlation; Distance; Hybrid	The adjacency matrix combines Pearson correlation and spatial distance: correlation below 0.5 gives no connection; correlation above 0.5 uses reciprocal node distance; diagonal/self values are 1.
S30 Zhang et al. (2024)	Air-quality micro-monitoring station	PMPre-Graph connects stations according to cosine similarity of the PM2.5/SO2/CO feature vectors; similarity is thresholded by β=0.8.	Weighted — similarity-based Gaussian-style weight for pairs satisfying the cosine-similarity threshold	Undirected	Time-evolving/statistical correlation graph	Correlation/statistical similarity; Hybrid pollutant-feature similarity	Feature vectors combine PM2.5, SO2 and CO. The adjacency weight uses cosine similarity sim(Si,Sj); the selected threshold is β=0.8, and σ is the standard deviation of the eigenvector. The authors construct correlation graphs intended to represent time-evolving cross-station PM2.5 relationships.
S31 Ma et al. (2022)	Heterogeneous monitoring point/station	All monitoring points are connected in a weighted graph whose pairwise weights are based on Gaussian spatial similarity of inter-point distance after RBF fusion	Weighted — Gaussian spatial-similarity weights	NR	Static fully connected geographical graph	Distance; Physical/geographical topology	RBF interpolation/fusion aligns heterogeneous sources according to monitoring-point coordinates. A 53×53 weighted adjacency matrix is then constructed using a Gaussian similarity function of spatial distance.
S32 Gu et al. (2025)	Pollutant monitoring station and integrated meteorological station within target-specific subgraphs	Stations are grouped geographically into target-specific subgraphs; initial spatial relationships are fused with evolving wind patterns and pollutant-transport information, and attention learns inter-station influence.	Weighted — graph edge weights are dynamically adjusted according to wind-driven transport and learned attention	NR	Dynamic hybrid graph; target-specific multi-subgraph/station-grouped graph	Distance; Meteorology; Wind/transport; Learned attention; Dynamic; Hybrid	Pollutant stations are grouped by K-means and nearby meteorological stations are integrated. Each target station receives a pruned local subgraph. Dynamic wind information is fused into edge weights to reflect time-varying inter-station pollutant transport; the graph-based prediction module then learns edge/node representations and attention weights.

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Supplementary Table S5 (continued)

Study	Graph node	Graph edge	Weighting	Direction	Graph type	Graph source	Construction details
S33 Zhang et al. (2024)	Air-pollutant monitoring station	Three spatial graph views are used: a distance graph, an adjacency/neighborhood graph, and a functional graph capturing inter-station relationships; attention fuses the graph views dynamically.	Mixed/weighted multi-graph — graph-specific weights plus learned attention fusion	Distance graph explicitly undirected; direction of the final fused attention graph NR	Multi-graph; adaptive/dynamic attention-fused graph	Distance; Neighborhood; Correlation/functional similarity; Multi-graph; Learned attention; Hybrid	MGAtt uses three subgraphs at different spatial scales: distance graph GD, adjacency/neighborhood graph GN, and functional graph GF. The graph views are combined with multi-graph attention so their relative contribution can vary by context before LSTM temporal modelling.
S34 Zhao et al. (2025)	PM2.5 monitoring station	A static directed geographical graph has trainable edge elements initialized by reciprocal distance; a dynamic directed wind-field graph is recomputed over time; the two are combined into a semi-dynamic directed graph.	Weighted — learned static geographical matrix combined with computed wind-field matrix	Directed	Semi-dynamic hybrid graph: static learned geographical graph + dynamic wind-field graph	Distance; Wind/transport; Meteorology; Learned; Adaptive; Dynamic; Hybrid	The static adjacency Ag is learned end-to-end; aij is initialized as reciprocal distance and values below an unspecified threshold are zeroed, with aij≠aji explicitly allowed. The dynamic Atw is calculated from ERA5 gridded wind using u/v components, wind direction along station-to-station paths and distance penalties. The two matrices are combined for dynamic graph convolution.
S35 Zhang et al. (2025)	City	An initial binary distance graph connects cities within ζ=200 km; learned city/support embeddings and vector-product relationships produce multi-granularity spatiotemporal graph structures, followed by graph attention.	Mixed — initial static graph is binary; learned spatiotemporal relationships/attention are weighted	Directed/asymmetric learned spatiotemporal relationships; initial distance graph is undirected	Multi-granularity adaptive spatiotemporal graph; hybrid static + learned graph	Distance; Learned; Adaptive; Multi-graph/multi-granularity; Hybrid	A static binary geographic graph uses a 200 km threshold. For each temporal granularity, trainable/support embeddings and vector products generate spatiotemporal relationship matrices; GAT then extracts spatial dependencies at each granularity before multi-scale feature fusion.
S36 Ye et al. (2025)	City	IPS first filters surrounding cities by instantaneous phase synchronization with the target city. The GNN then learns a target-oriented spatial impact matrix between the retained cities and the target across time.	Weighted — learned spatial-impact values	Directed/target-oriented spatial impacts	Learned/adaptive target-specific graph after synchronization-based filtering	Correlation/synchronization; Meteorology/wind; Geography; Distance; Elevation; Population; Learned; Adaptive; Hybrid	For each target city, the IPS estimator computes instantaneous phase synchronization with all other cities and retains cities whose synchronization score exceeds threshold β. A supervised GNN then learns the spatial-impact matrix among the target and retained surrounding cities; the resulting impact sequence is passed to the Transformer module.
S37 Velez-Serrano and Alvaro-Meca (2026)	Municipality	Delaunay triangulation defines the geographical neighborhood; edge weights incorporate planar geographic distance, altitude difference and population-related weighting.	Weighted — adaptive/stationary geographical edge weights combine distance, altitude and population terms	NR	Static/stationary adaptive geographical graph	Physical/geographical topology; Distance; Elevation; Population; Delaunay triangulation; Adaptive	A Delaunay triangulation is constructed over municipal coordinates. Edge weights combine normalized geographical distance and altitude difference with a population-related factor; the adaptive graph is then used by the spatial GCN layers for high-resolution municipal forecasting/extrapolation.
S38 Dong et al. (2024)	Air-quality monitoring station	A weighted directed station graph is constructed using Gaussian distance-based weights between stations; edges beyond a distance threshold are set to zero.	Weighted — Gaussian distance-decay weight	Directed	Static physical/geographical graph with uncertainty-guided graph convolution	Distance; Physical/geographical topology	The adjacency uses a Gaussian distance kernel W=exp(-dist(si,sj)^2/σ^2) for station pairs within threshold ξ, otherwise 0. AirNN then treats graph-convolution parameters probabilistically through variational inference; the underlying adjacency itself is distance-based/static.
S39 Dun et al. (2022)	Air-quality monitoring station	Dynamic adjacency combines PM2.5 temporal correlation from sliding-window grey relational analysis with station-distance information.	Weighted	Undirected	Dynamic hybrid correlation–distance graph	Correlation; Distance; Dynamic; Hybrid	A time-varying correlation matrix is generated from sliding-window grey relation analysis of PM2.5 sequences and combined with inverse-distance information computed from station coordinates to form the dynamic adjacency used by graph convolution.
S40 Tariq et al. (2023)	Air-quality monitoring station	Distance-adaptive graph construction samples/links nearby nodes using station-distance information and learns adaptive node relationships through trainable node dictionaries/adjacency.	Weighted — learned adaptive graph weights	NR	Adaptive/learned distance-aware graph	Distance; Learned; Adaptive; Hybrid	The DAGG module uses node sampling based on a station-distance matrix and learns adaptive graph relationships through node embeddings/dictionaries; the learned graph is embedded in gated recurrent graph-convolution cells.
S41 Wang et al. (2024)	City	Cities are connected if distance is within 200 km and the maximum terrain obstacle/altitude difference condition is below 1200 m.	Mixed — binary physical adjacency plus learned GAT attention weights	Initial physical graph is based on symmetric distance/terrain criteria; learned attention direction NR	Static physical/geographical graph with learned attention; hybrid	Distance; Physical/geographical topology; Topography; Meteorology; Learned attention; Hybrid	Adjacency Aij is defined using Heaviside functions of inter-city distance dij and maximum terrain obstacle mij, with thresholds dθ=200 km and mθ=1200 m. Meteorological variables are node attributes and GAT learns connection weights.
S42 Zhang et al. (2024)	City	Candidate city pairs are connected using geographical distance and terrain constraints; transport-relevant edge attributes additionally encode distance, source wind speed/direction, source-to-target direction, advection/turbulence coefficient and learned attention.	Mixed — binary physical adjacency plus continuous edge attributes/attention weights	Directed transport/edge representation	Static physical topology with meteorological edge attributes and learned graph attention; hybrid	Distance; Topography; Meteorology; Wind/transport; Learned attention; Hybrid	Physical connectivity uses a 300 km distance threshold and a terrain/mountain constraint involving mountains above 1200 m. Edge attributes include distance, wind speed/direction, source-target direction, advection coefficient, and learned attention.
S43 Liu et al. (2024)	Air-quality monitoring station	Two static graph variants are evaluated: geographical-proximity graph using a Gaussian distance kernel and historical-PM2.5 correlation graph using thresholded correlation.	Weighted	NR	Static graph; two alternative graph variants	Distance; Correlation	Geographical adjacency uses a Gaussian kernel Aij=exp(-dij²/σ²) subject to a tunable threshold ε. The alternative graph uses historical PM2.5 correlation with a tunable threshold η. Thresholds are tuned over 0.0–0.9.
S44 Mohammadzadeh et al. (2024)	EPA monitoring site/city location	A graph adjacency matrix is used by the GCN to aggregate spatial information across sites; the specific rule for constructing the adjacency matrix is NR.	NR	NR	Static graph; exact dependency source NR	NR	The supplied full text describes graph convolution using a normalized adjacency matrix with self-loops but does not report the study-specific rule used to define station-to-station edges.
S45 Liao et al. (2024)	PM2.5 monitoring location	Directed graph edges represent spatial interactions/transport among monitoring locations using source-node wind speed/direction and pairwise direction/distance; elevation differences are also included as domain knowledge.	Unclear from full text — edges carry transport/geographical attributes, but a single scalar adjacency-weight rule is not explicitly reported	Directed	Directed domain-informed spatiotemporal graph with time-varying meteorological edge attributes	Distance; Wind/transport; Meteorology; Elevation/topography; Hybrid	The adopted PM2.5-GNN represents monitoring sites as nodes and uses directed source-to-target spatial relationships with wind, relative direction, distance and elevation information. Fire influence is added through wind inverse-distance-weighted FRP features.
S46 Han et al. (2025)	Unclear from full text — the methods describe graph nodes generically as geographic locations or climate variables, but do not state the exact study-specific node definition	Unclear from full text — the methods refer generically to connections/correlations but do not report a study-specific edge-construction rule	NR	NR	NR	NR	Unclear from full text
S47 Huang et al. (2025)	City / monitoring location	A binary geographical graph connects locations when inter-city distance is within 400 km and the path does not contain three or more mountains above 1200 m.	Unweighted/binary initial adjacency	NR	Static physical/geographical graph	Distance; Physical/geographical topology; Topography	Adjacency is defined by distance and terrain using threshold conditions: distance ≤400 km and fewer than three intervening mountains with altitude >1200 m; otherwise no edge.

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Supplementary Table S5 (continued)

Study	Graph node	Graph edge	Weighting	Direction	Graph type	Graph source	Construction details
S48 Wang et al. (2024)	Air-quality monitoring station	Weighted station graph based on Haversine inter-site distance and a Gaussian distance-decay connectivity function with sparsity threshold ξ.	Weighted — Gaussian distance-decay edge weights	Undirected by distance-based station relationship	Static physical/geographical graph	Distance; Physical/geographical topology	Eight station nodes are linked using Haversine distance dij. Edge weight Aij is based on exp(-dij²/σ²) subject to threshold ξ, producing a sparse weighted adjacency. Node features combine air quality, meteorology and POI variables.
S49 Dimri et al. (2024)	Air-quality monitoring station	No predefined physical graph is used. For every node pair, attention coefficient eij is computed and a learned connection strength Aij is generated by a residual two-layer MLP; edges are retained where Aij>0.6.	Weighted — softmax-normalized learned attention coefficients	Directed/adaptive influence j→i	Adaptive learned graph	Learned; Adaptive	AGATNet dynamically learns the graph from node features instead of using distance/domain rules. Aij=F(xi,xj)=σ(eij + W2(σ(W1(eij)))); edges with Aij≤0.6 are pruned; retained attention coefficients are softmax-normalized.
S50 Qi et al. (2024)	City in the main experiment; monitoring site in the Shenyang fine-scale experiment	Initial adjacency Aij=1 when distance(vi,vj)≤DISTANCE and 0 otherwise; GAT then learns attention coefficients over connected neighbors.	Mixed — binary distance-based initial adjacency plus learned GAT attention weights	NR	Static geographical graph with learned graph attention	Distance; Learned attention; Hybrid	A distance-threshold adjacency is constructed among nodes. Five-fold cross-validation is used to select DISTANCE and temporal-window/period hyperparameters; the selected distance threshold is 300 (unit reported elsewhere in the paper as km for DISTANCE).
S51 Pan et al. (2024)	City	A directed city graph uses a 400 km distance threshold and 1200 m mountain-height threshold; connected city pairs are represented with two-way directed edges. Edge attributes include distance and angle direction, while wind speed/direction are node meteorological inputs used by the GNN.	Mixed — binary physical connectivity plus learned attention/message weights	Directed with bidirectional edges for connected city pairs	Static physical graph with learned graph attention	Distance; Topography; Meteorology; Wind/transport; Learned attention; Hybrid	An edge exists if city distance<400 km and the connecting path does not exceed the 1200 m mountain-range threshold. Distance and angle direction are edge attributes. GARNN uses pairwise city hidden features and edge attributes in graph message passing and attention re-weighting.
S52 Jin et al. (2023)	Beijing administrative district	Each node is connected to its N nearest adjacent districts according to geographical distance; Bayesian optimization selected edges_per_node=3.	NR	NR	Static k-nearest-neighbour geographical graph	Distance; Physical/geographical topology	Graph nodes are the 16 Beijing districts. Distances are calculated from district longitude/latitude; each node is assigned its edges_per_node nearest adjacent districts. Bayesian optimization selected edges_per_node=3.
S53 Zhang et al. (2022)	Time moment in the constructed temporal graph	Graph structure is constructed from original PM2.5 time-series moments; the developed graph-attention mechanism incorporates first-order backward temporal differences between adjacent moments.	Weighted — attention weights	NR	Temporal graph constructed from time-series moments	Temporal difference; Learned attention; Other	TDGTN converts time-series observations at different moments into graph-structured data and embeds first-order backward differences between adjacent moments into a graph attention mechanism used in Transformer encoders/decoders.
S54 Zeng et al. (2022)	Monitoring station	Spatial diffusion cost between stations combines straight-line distance and wind direction using a simplified Gaussian-diffusion formulation.	Weighted — wind-direction/distance diffusion-cost adjacency	Directed transport cost i→j	Dynamic meteorology-informed graph	Distance; Meteorology; Wind/transport; Dynamic; Hybrid	For edge Eij, the simplified cost is cost(Eij)=[2×F(Di,DM)]×Lij, where Di is wind-direction azimuth, DM is the edge azimuth, Lij is inter-station distance, and F returns the absolute azimuth difference. This augments conventional distance-only adjacency with wind effects.
S55 Lu et al. (2025)	Air-quality monitoring station; higher-level learned station-group nodes are additionally formed	Geographical adjacency A_geo connects stations within 18 km; adaptive adjacency A_adp=SoftMax(ReLU(E1E2^T)) captures latent station relations. Learned station groups form dynamic group-to-group edges modulated by temporal encoding.	Mixed — binary geographical adjacency plus weighted learned adaptive adjacency, attention, and soft station-group memberships	NR	Dynamic adaptive multi-scale hierarchical graph	Distance; Learned; Adaptive; Dynamic; Hybrid	MSDGNN combines a fixed 18-km geographic graph with an adaptive adjacency learned from node embeddings, station-group soft assignment matrix S, dynamic inter-group temporal edge attributes, multi-head attention, and Chebyshev graph convolution.
S56 Abbasi et al. (2025)	Air-quality monitoring station	A separate pollutant graph is constructed for each selected pollutant; two stations are connected with Aij=1 only when i≠j and both AQMSs belong to the same pollutant-specific cluster, otherwise Aij=0.	Unweighted/binary	Undirected	Static fixed multi-graph / clustered graph	Correlation/statistical time-series clustering; Multi-graph	Time-frequency transformation + PCA + agglomerative hierarchical clustering groups stations separately for each pollutant. Selected pollutants with meaningful Copula dependence and spatial clustering (PM2.5, PM10, NO2) each receive a fixed binary within-cluster adjacency.
S57 Guan et al. (2025)	10-km raster grid cell/grid point	All grid nodes are related through a weighted distance adjacency based on Haversine great-circle distance; edges are retained when exp(-dij/σ²)≥ε.	Weighted — Wij=exp(-dij/σ²) subject to sparsity threshold ε; modified self-loop/neighbor weighting is used in the improved GCN	Undirected	Static physical/geographical raster graph	Distance; Physical/geographical topology	Haversine distance is computed between grid points. Initial adjacency uses σ²=10 and ε=0.5. The improved GCN uses self-loop redistribution parameter p≈0.91 and K-hop propagation; validation selected K=5.
S58 Chen et al. (2024)	Air-quality monitoring station	Initial adjacency is based on inverse Euclidean distance between monitoring stations; the adjacency is then adaptively updated using meteorological and POI-related information.	Weighted — initial inverse-distance adjacency combined with learned adaptive relationships	NR	Adaptive learned hybrid graph	Distance; Meteorology; POI; Learned; Adaptive; Hybrid	The model first constructs a prior adjacency from station distance and then learns an adaptive adjacency matrix using trainable graph relations informed by meteorological and POI information, reducing dependence on a fixed prior graph.
S59 Chauhan et al. (2026)	Air-quality monitoring station	Two monitoring stations are connected when their Haversine distance is less than 30 km; graph attention is then used for spatial aggregation.	Mixed — binary geographical adjacency with learned GAT attention coefficients	NR	Static geographical graph with learned attention	Distance; Learned attention; Hybrid	A fixed station graph is created using a 30 km distance threshold. GAT operates on this graph, while a separate feature-attention module learns the importance of environmental variables.
S60 Li et al. (2023)	Air-quality monitoring station	Two stations are connected when their Spearman correlation coefficient is not less than 0.9.	Unweighted/binary adjacency	Undirected	Static correlation/statistical graph	Correlation	A station correlation network is constructed with Spearman rank correlation. An edge is retained when the correlation coefficient is ≥0.9. Chebyshev graph convolution (K=2) is then applied.
S61 Mirzaei and Torghabeh (2026)	Air-quality monitoring station	Five stations form a fully connected directed graph including self-loops. Each pair has meteorology-aware/data-driven edge features built from distance, bearing, wind vectors and Neural-CDE pollutant states.	Weighted — learned positive edge weights/attention from edge-feature MLP and GAT	Directed	Adaptive/dynamic meteorology-aware fully connected graph	Distance; Meteorology; Wind/transport; Learned; Adaptive; Dynamic; Hybrid	For every directed station pair, the model concatenates normalized distance, bearing components, source wind-vector components, and CDE-derived source/target pollutant states. A learned edge MLP and GAT then determine spatial interaction strengths.
S62 BADOUCH and BELHOUCINE (2026)	Air-quality monitoring station	Two alternative graphs are tested: distance topology connects each station to its k=5 nearest stations using Haversine distance; correlation topology connects each station to its k highest PM2.5-correlated stations, with correlations computed only on the training split.	Weighted — distance graph uses wij=1/dij; correlation graph uses correlation-derived neighborhood selection/weights as reported	NR	Static graph; alternative distance and correlation graph variants	Distance; Correlation	Distance graph: k=5 nearest neighbours by Haversine distance, with inverse-distance edge weight. Correlation graph: Pearson PM2.5 correlations computed exclusively on the training split to avoid leakage; k values 3, 5, 7 and 10 are evaluated.

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Supplementary Table S5 (continued)

Study	Graph node	Graph edge	Weighting	Direction	Graph type	Graph source	Construction details
S63 Huang et al. (2025)	City	The framework combines a dynamically learned global spatiotemporal graph with a local geographically/transport-informed graph using edge attributes related to distance and wind transport.	Weighted	Directed transport relationships in local graph; global learned relationship direction as modelled by attention	Dynamic hybrid global–local graph	Distance; Wind/transport; Meteorology; Learned; Adaptive; Dynamic; Hybrid	A global BSTA module adaptively learns dynamic spatiotemporal dependencies, while the local GESM module models physically meaningful local city relationships using geographical and wind-transport edge features; outputs are fused for forecasting.
S64 Pei et al. (2022)	Multivariate time-series variable/pollutant feature rather than a spatial monitoring station	A latent correlation graph among multivariate time-series variables is learned using self-attention/correlation relationships before graph convolution.	Weighted — learned attention/correlation matrix	NR	Learned/adaptive latent correlation graph	Correlation; Learned; Adaptive	The MtemGNN component builds a learned relationship matrix among multivariate pollutant/AQI time series through a self-attention mechanism and applies graph convolution to extract cross-variable temporal dependencies.
S65 Yu et al. (2026)	Air-pollution monitoring station	Base spatial dependencies are restricted to station pairs within a 200 km radius; reciprocal-distance weights are used and multi-head GAT learns state-dependent station influence.	Weighted — inverse-distance base weights plus learned attention weights	Mixed — reciprocal-distance base graph with directional attention/influence patterns	Adaptive/dynamic attention over a static geographical topology	Distance; Learned; Adaptive; Dynamic; Hybrid	Stations within 200 km form the physical spatial graph; base weights use inverse distance. Multi-head GAT learns context-dependent spatial interaction strengths, while a Koopman module models temporally evolving latent dynamics.
S66 Wang et al. (2024)	Monitoring station	A weighted undirected station graph combines Gaussian distance similarity and Pearson pollutant-correlation similarity; edges below threshold θ are removed.	Weighted — hybrid distance/correlation edge weight controlled by β	Undirected	Static hybrid/multi-graph construction across VMD components	Distance; Correlation; Hybrid; Multi-graph	Edge similarity combines a distance-decay term and Pearson-correlation term using mixing coefficient β and threshold θ. VMD decomposes the signal into components, and GAT operates on graph representations before BiLSTM temporal learning.
S67 Zhao (2025)	Air-quality monitoring station	Binary geographic adjacency: two stations are connected when their distance is below the average geographical distance threshold; self-connections are added in graph convolution.	Unweighted/binary	NR	Static physical/geographical graph	Distance; Physical/geographical topology	The station network uses a binary distance-threshold adjacency based on whether pairwise station distance is below the average inter-station geographical distance. ST-GCN then jointly models spatial and temporal structure.
S68 Zeng et al. (2021)	Monitoring station	Directed weighted edge cost models pollutant diffusion from station A to B using wind-direction azimuths and inter-station distance; pollution-source influence around each node is additionally fused into the adjacency.	Weighted — diffusion cost from wind direction and distance, fused with pollution-source influence coefficient	Directed	Dynamic meteorology/industrial-source-informed graph	Distance; Meteorology; Wind/transport; Industrial/emission; Dynamic; Hybrid	The paper defines cost(EAB)=[F(DA,DM)+F(DB,DM)]×LAB and, because wind direction is assumed uniform across the industrial park at the same time, simplifies it to cost(EAB)=[2×F(DA,DM)]×LAB. Pollution sources within radius S=0.2 km are converted to an influence coefficient c (default r=3) and fused into the adjacency matrix.
S69 Wang et al. (2022)	City	A binary physical adjacency connects two cities when distance dij<300 km and highest intervening mountain elevation hij<1200 m.	Mixed — binary physical adjacency with learned GAT attention weights	NR	Static physical graph with learned attention	Distance; Physical/geographical topology; Topography; Meteorology; Learned attention; Hybrid	Aij=1 when dij<dk and hij<hk, otherwise 0, with dk=300 km and hk=1200 m. Seven meteorological variables are used as city-node attributes and GAT learns spatial attention over connected cities.

Notes. Values reproduce the supplied 69-study extraction. NR = not reported; uncertain or partially described properties are not inferred. S01–S69 are row identifiers (full titles and DOIs in the companion study registry). Graph categories and relation sources can overlap. An edge reflecting correlation, attention, or learned similarity is not automatically a physically validated atmospheric transport connection.